

COMP1008 Part II: Machine Learning

- **Machine learning I:** Intro (last week)
 - **Machine learning II:** [Process & Data Processing, Un-supervised Learning](#) (this week)
 - **Machine learning III:** Supervised learning
 - **Machine learning IV:** ANN & extensions
 - Computing session 1: [NumPy and Pandas](#)
- 

Machine Learning : (Main) Types of Tasks

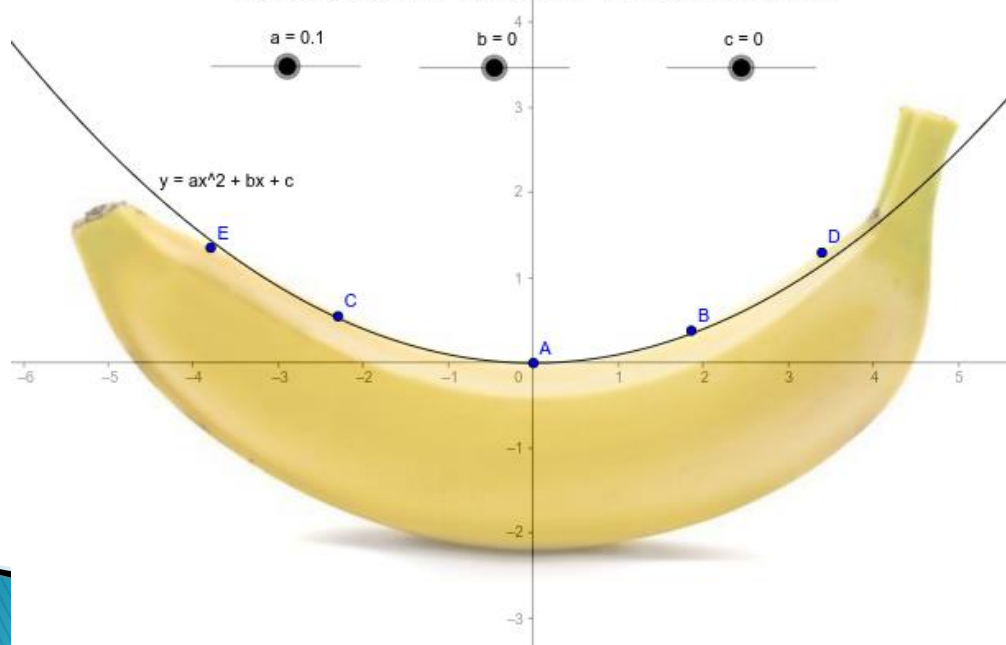
Quadratic equation in standard form

$$ax^2 + bx + c = 0$$

Quadratic Formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

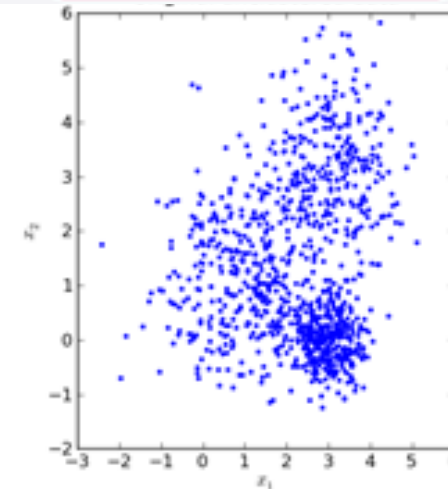
Quadratic Function on a Banana



Machine learning

Supervised learning

Unsupervised learning



Machine Learning : (Main) Types of Tasks

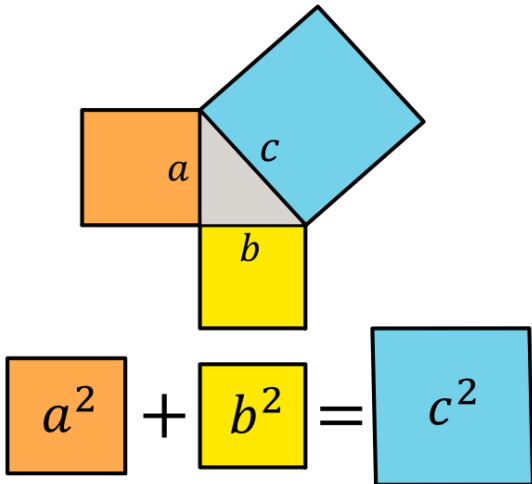
Quadratic equation in standard form

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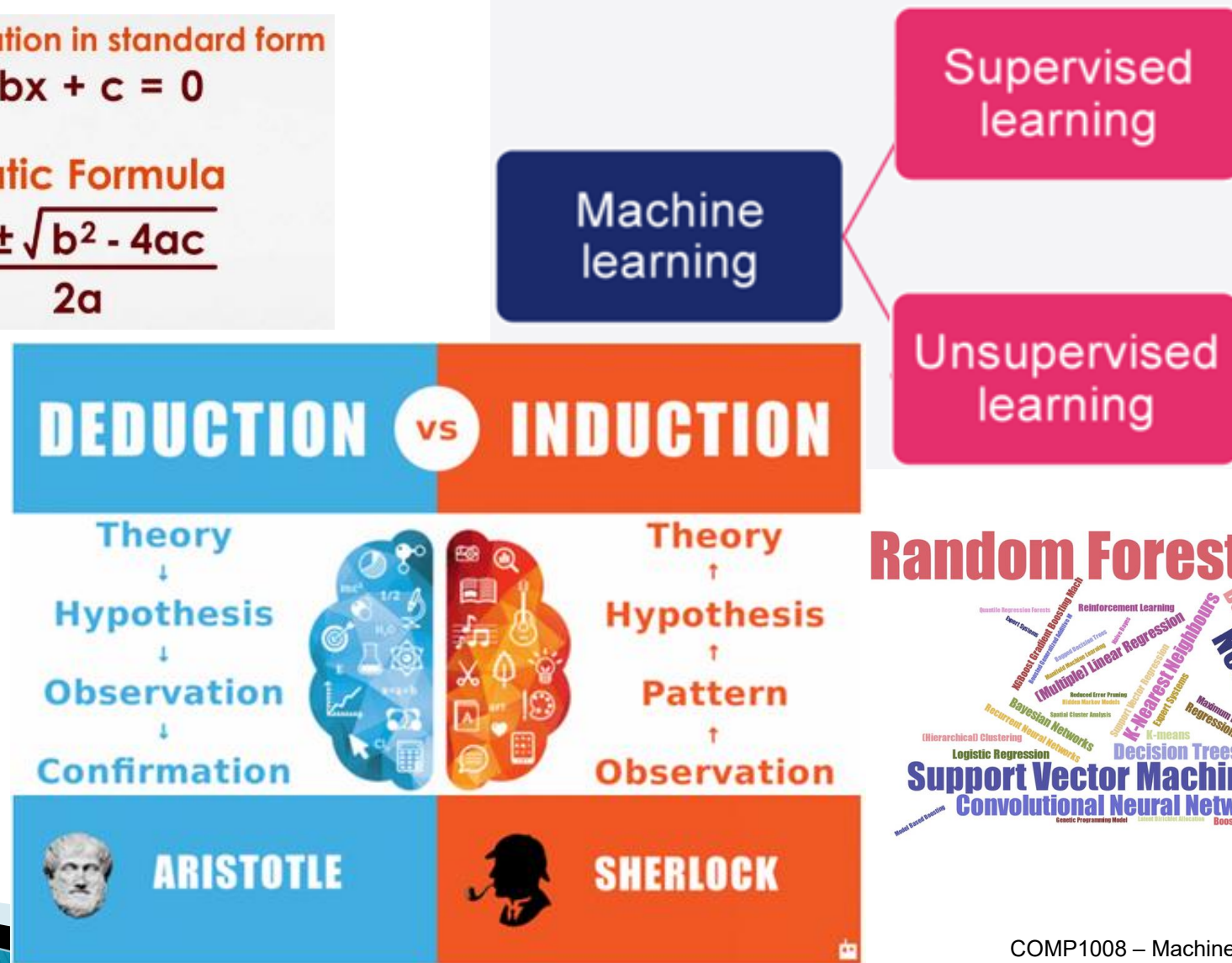
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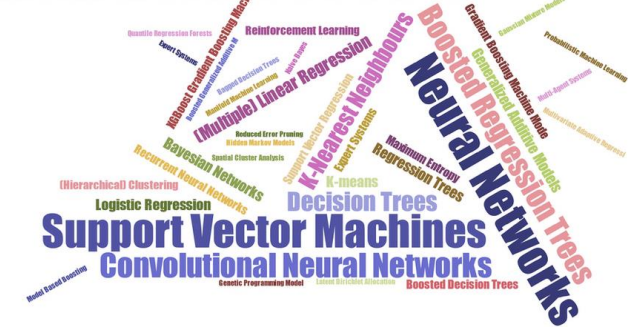
PYTHAGOREAN THEOREM



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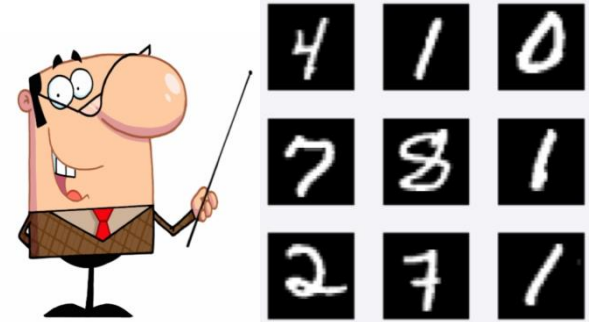
Random Forests



Machine Learning : (Main) Types of Tasks

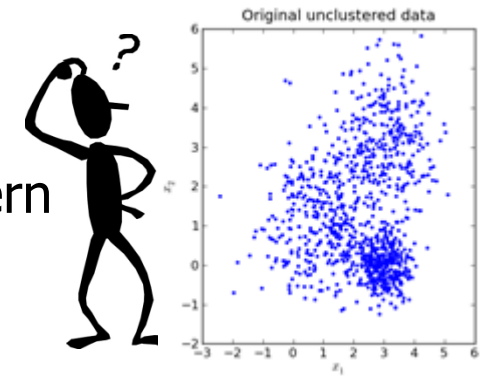
► Supervised learning

- Task: learns from **labelled** training data (each sample/instance /data point consists of an input and a **desired** output)
- Goal: find/train a model to match output to input
- Algorithms: **Linear regression, Decision tree, ANN, etc.**
- Examples: classifications, predictions



► Unsupervised learning

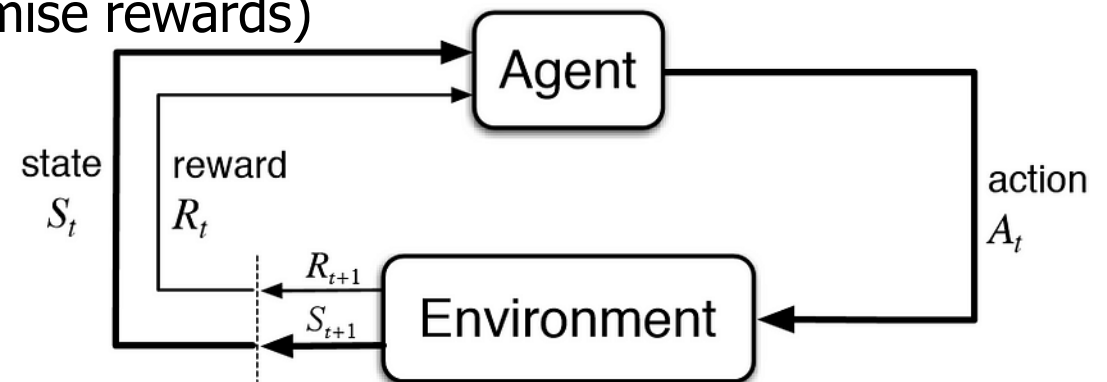
- Task: learns from **unlabelled** data to describe hidden structure/pattern
- Goal: task dependent (e.g. group data points, find the distribution)
- Algorithms: **K-means, etc.**
- Examples: clustering/grouping, association rules



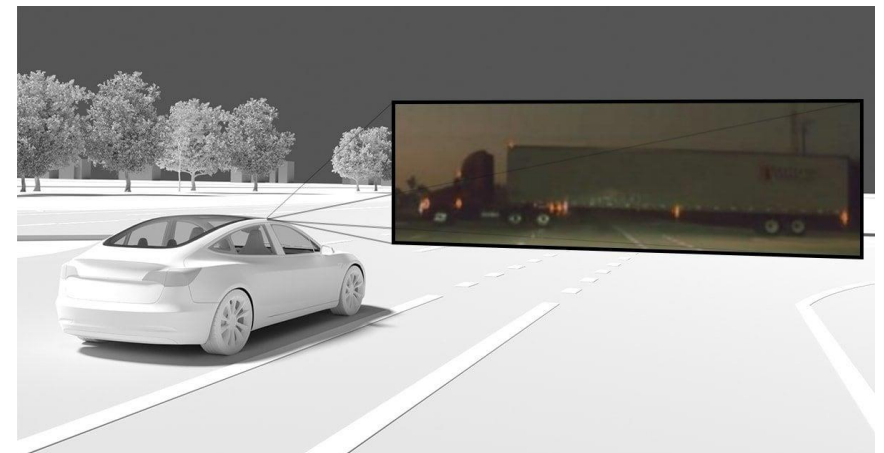
Machine Learning : (Main) Types of Tasks

► Reinforcement learning

- Learns over **time** via trial and error using feedback / **awards from actions**
- Goal: learn a **policy** to make decisions (to maximise rewards)
- Algorithms: Deep Neural Networks, Q-Learning, Deep Q-networks, etc.
- Examples: self-driving cars, robotics, game playing, etc.
- (Not covered in COMP1008)

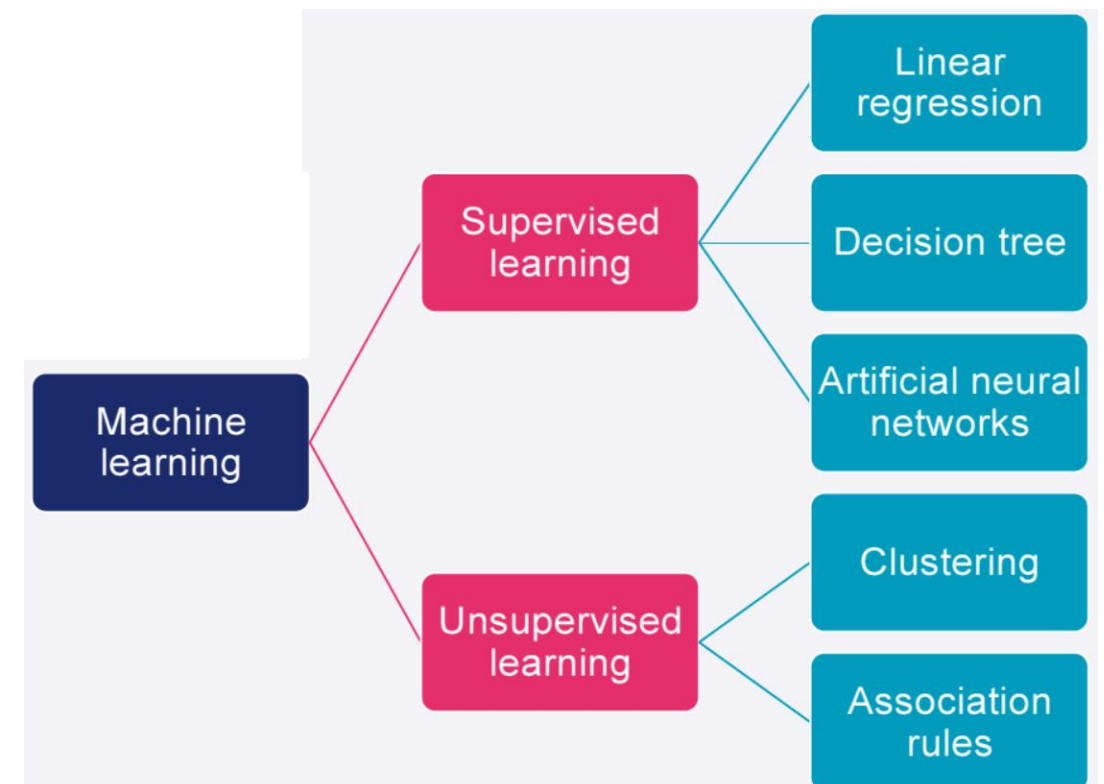
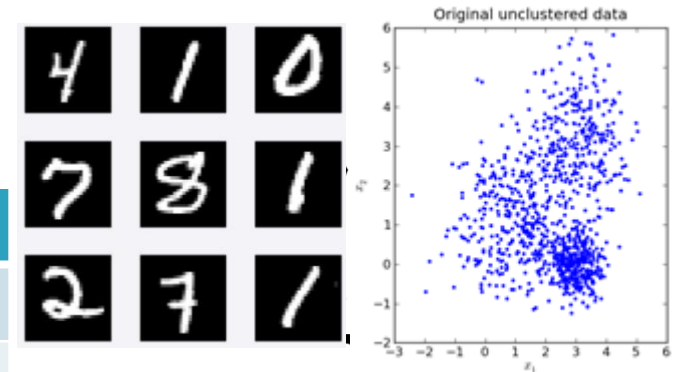


AI	COMP1008	CS Course
Bottom-up (Inductive)	Machine Learning	COMP3009



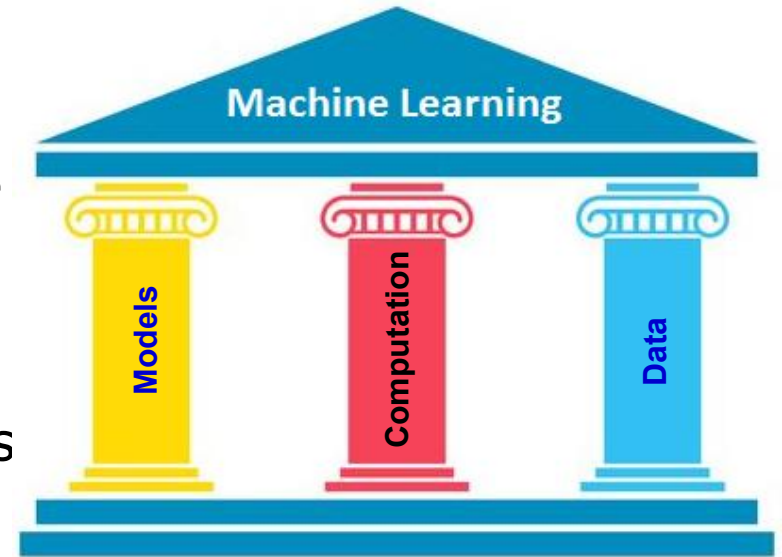
Machine Learning : (Main) Types of Tasks

Supervised	Unsupervised
$y = F(x)$: relationship between y and x	$y = ?$: usually no function
D : labeled training set	D : unlabeled data set



Machine Learning : The Three Pillars

- ▶ Three pillars of machine learning
 - **Models:** AI modules in the school
 - Early success but more breakthroughs the last decade
 - Most recent success due to deep learning
 - **Computation**
 - Not the focus in COMP1008
 - Less an issue(?) hopefully with scientific developments
 - **Data:** often key for successful machine learning!
 - A key factor for the different LLM behaviours
 - Led to some of the breakthroughs
 - A bottleneck but may be resolved(?) with more research breakthroughs

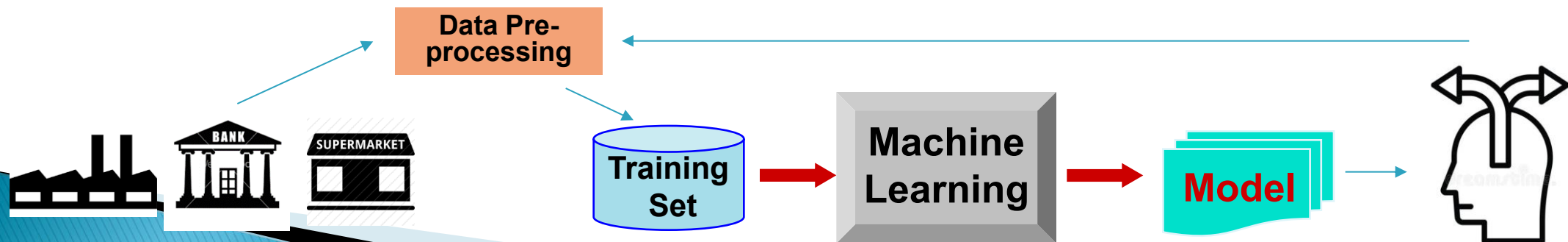
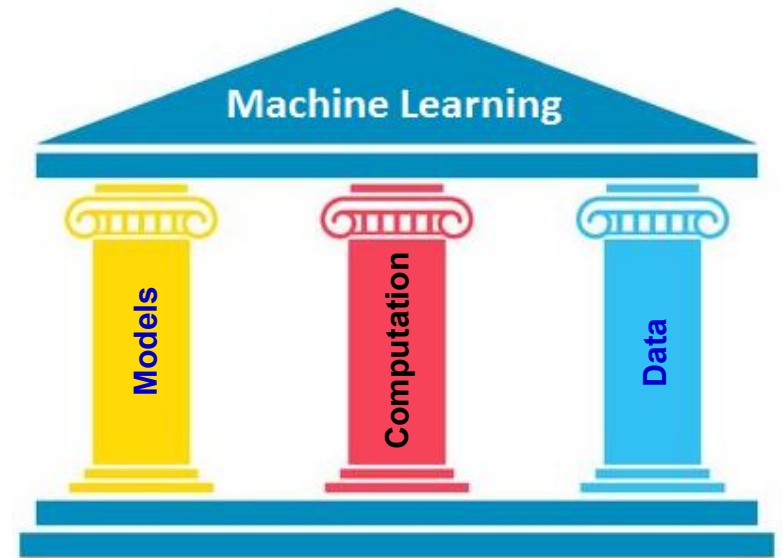


Machine Learning : Process (un-supervised learning)

- Prepare the **training data**



Learning the parameters of the model



Machine Learning : Process (supervised learning)

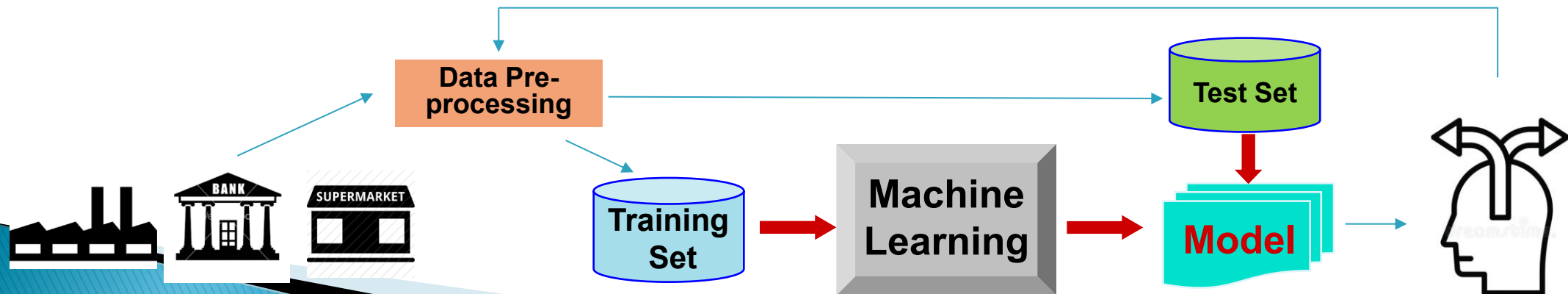
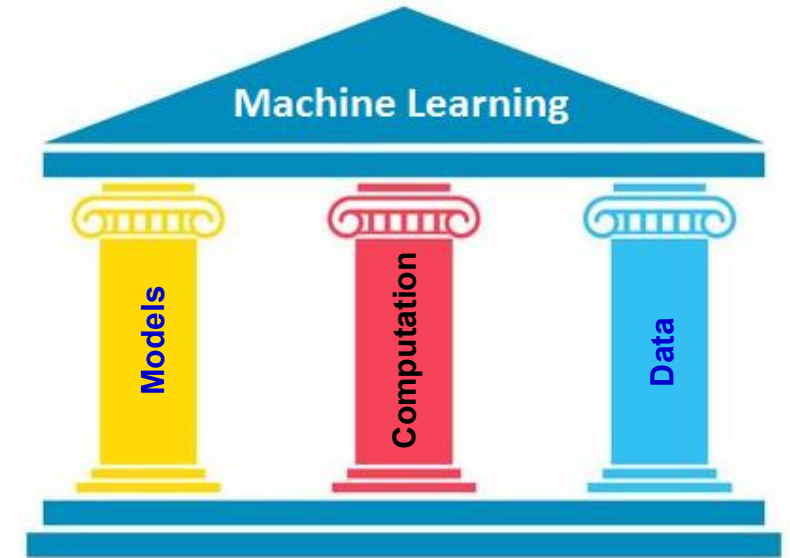
- ▶ Partition the total dataset into **two subsets**
 - 70% training vs. 30% testing



Learning the parameters of the model

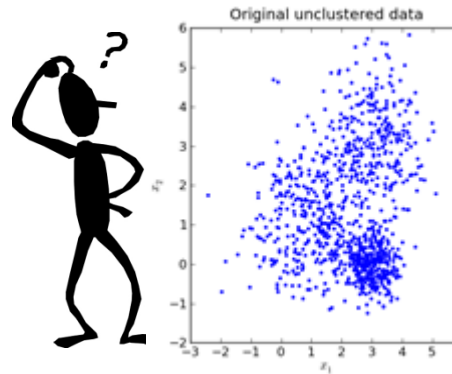


How to **generalize** to new data, i.e. overfitting
Unbiased estimate of how well the model works



Machine Learning : Data Pre-processing

- ▶ Unsupervised learning
- ▶ Understand the problem
 - Features
 - Which algorithms/approaches?

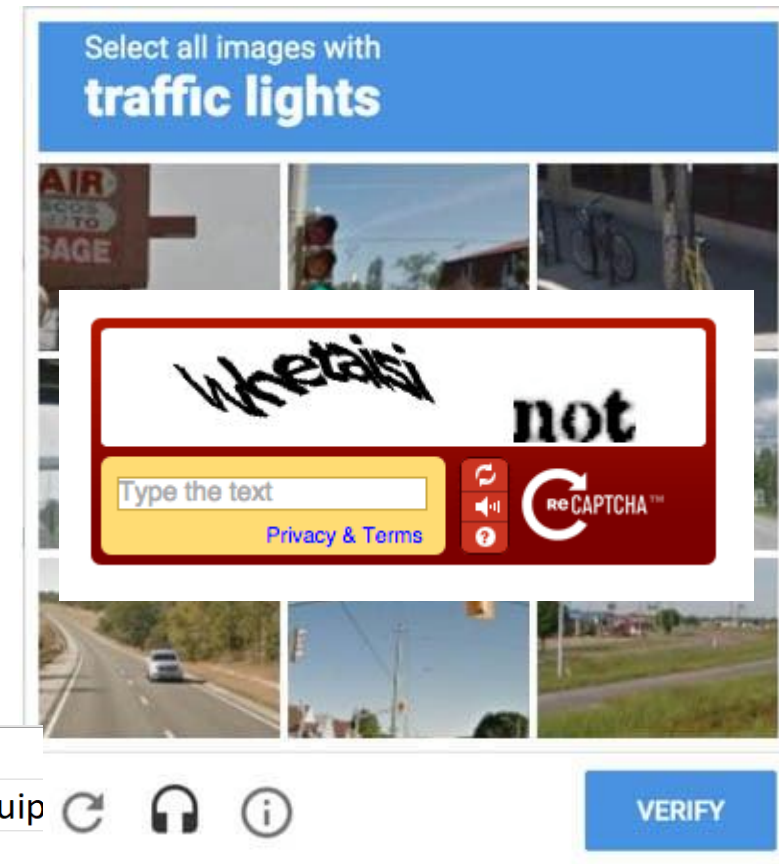


ID	Age	Sex	Job	Housing	SavingAcc	CheckAcco	CreditAmc	Duratio	Purpose
995	31	female	1	own	little	NA	1736	12	furniture/equip
996	40	male	3	own	little	little	3857	30	car
10	25	female	2	rent	little	moderate	1295	12	car
17	25	male	2	own	NA	little	8072	30	business
999	27	male	2	own	moderate	moderate	4576	45	car

Features/Attributes

31, male, 1, rent, moderate, little, 2836, 20, car

?



Samples/
Observations/
Instances

Machine Learning : Data Pre-processing

- ▶ Supervised learning
- ▶ Understand the problem
 - Features and Labels
 - Which models/approaches?

ID	Age	Sex	Job	Housing	SavingAcc	CheckAcco	CreditAm	Duration	Purpose	Risk
0	67	male	2	own	NA	little	1169	6	radio/TV	good
1	22	female	2	own	little	moderate	5951	48	radio/TV	bad
2	49	male	1	own	little	NA	2096	12	education	good
3	45	male	2	free	little	little	7882	42	furniture/e	good
4	53	male	2	free	little	little	4870	24	car	bad
5	35	male	1	free	NA	NA	9055	36	education	good

Features/Attributes

Labels/Targets

Samples/
Observations/
Instances



Machine Learning : Data Pre-processing

- ▶ Process the data into a suitable format for machine learning (time consuming!)
 - **pandas DataFrame** library
 - Improve model accuracy
 - Reduce noises and bias

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4	53	male	2	free	little	little	4870	24	car	bad
5	35	male	1	free	NA	NA	9055	36	education	good
6	53	male	2	own	quite rich	NA	2835	24	furniture/eq	good
7	35	male	3	rent	little	moderate	6948	36	car	good
8	61	male	1	own	rich	NA	3059	12	radio/TV	good
9	28	male	3	own	little	moderate	5234	30	car	bad

```
import pandas as pd
```

```
df = pd.read_csv("my_csv_file.csv")
```

how to read an excel file?

```
df['Age'].mean()
```

Computing sessions 1&2

Machine Learning : Data Pre-processing

- ▶ Process the data into a suitable format for machine learning (time consuming!)
 - Data collection
 - Gather **relevant** data from sources
 - Data transformation
 - Encode **categorical** values
 - **Normalise** numerical data
 - Split data (training, testing)

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Machine Learning : Data Pre-processing

- ▶ Process the data into a suitable format for machine learning (time consuming!)
 - Data cleaning
 - Imputation: handle missing data

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df.isnull()

df.dropna() # remove the rows with missing data

df.replace(to_replace = np.nan, value = m) # replace the missing data with a value

df['SavingAcc'] = df['SavingAcc'].fillna("little") # more in the computing exercises

Computing sessions 2

Machine Learning : Data Pre-processing

- ▶ Process the data into a suitable format for machine learning (time consuming!)
 - Data cleaning
 - Imputation: handle missing data
 - Remove outliers, duplicates (bias)
 - Data visualisation: **matplotlib**
`import matplotlib.pyplot as plt`
`plt.scatter(df['Weight'], df['Height'])`
`plt.xlabel('Weight')`
`plt.ylabel('Height')`
`plt.title('Weight vs. Height: Men vs. Women')`

Computing sessions 2

